

Model dossier — B-L-SM

Status: reverse run failed. The blank-slate reconstruction aborted on an agent transport error (connection reset). Infrastructure, not physics.

This dossier is not a completed reverse-check review package. It collects what exists for this model so you can read the physics without opening the source files. Where a full review package does exist for a model, it is the REVIEW.pdf in that model's directory.

item	value
model	B-L-SM
chain status	passed the full chain
Lagrangian source	repair2/final.fr (re-validated)
Lagrangian terms found	9

Verbatim Lagrangian terms

Quoted unmodified from the .fr. These are the terms any reconstruction would have to account for.

LZpKin (:=)

```
Block[{mu,nu}, -1/4 FS[Zp,mu,nu] FS[Zp,mu,nu]]
```

LChi (:=)

```
Block[{mu,ii}, ExpandIndices[(del[HC[Chi],mu] - I*2*g1p*Zp[mu]*HC[Chi])*(del[Chi,mu] +  
  ↳ I*2*g1p*Zp[mu]*Chi) - muChi2*HC[Chi]*Chi - lambda2BL*(HC[Chi]*Chi)^2 -  
  ↳ lambda3BL*(Phibar[ii]*Phi[ii])*(HC[Chi]*Chi), FlavorExpand -> SU2D]]
```

LBLCurrent (:=)

```
Block[{mu}, ExpandIndices[-g1p*Zp[mu]*(1/3 QLbar.Ga[mu].QL + 1/3 uRbar.Ga[mu].uR + 1/3  
  ↳ dRbar.Ga[mu].dR - LLbar.Ga[mu].LL - lRbar.Ga[mu].lR - nuRbar.Ga[mu].nuR), FlavorExpand  
  ↳ -> {SU2D, Generation, Colour}]]
```

YMRRules (=)

```
{YMRMat[1,1] -> YMR1x1, YMRMat[1,2] -> YMR1x2, YMRMat[1,3] -> YMR1x3, YMRMat[2,1] -> YMR2x1,  
  ↳ YMRMat[2,2] -> YMR2x2, YMRMat[2,3] -> YMR2x3, YMRMat[3,1] -> YMR3x1, YMRMat[3,2] ->  
  ↳ YMR3x2, YMRMat[3,3] -> YMR3x3}
```

YNURules (=)

```
{YNUMat[1,1] -> YNU1x1, YNUMat[1,2] -> YNU1x2, YNUMat[1,3] -> YNU1x3, YNUMat[2,1] -> YNU2x1,  
  ↳ YNUMat[2,2] -> YNU2x2, YNUMat[2,3] -> YNU2x3, YNUMat[3,1] -> YNU3x1, YNUMat[3,2] ->  
  ↳ YNU3x2, YNUMat[3,3] -> YNU3x3}
```

LNuYukNonHC (:=)

```
Block[{sp,ff1,ff2}, ExpandIndices[-YNUMat[ff1,ff2] (LLbar[sp,1,ff1].nuR[sp,ff2] (Phibar[2] -  
  ↳ vev/Sqrt[2]) - LLbar[sp,2,ff1].nuR[sp,ff2] Phibar[1]) - 1/2 YMRMat[ff1,ff2]  
  ↳ anti[CC[nuR]][sp,ff1].nuR[sp,ff2] Chi, FlavorExpand -> Generation] /. YNURules /.  
  ↳ YMRRules]
```

LNuYuk (:=)

```
LNuYukNonHC + HC[LNuYukNonHC]
```

```

LScalarDiag (:=)
Block[{mu}, ca*sa*del[H,mu]*del[H2,mu] - ca*sa*muChi2*H*H2 - ca*sa*lambda3BL*vev^2*H*H2/2 +
  ↪ ca*lambda3BL*vev*xBL*H*H2 - 3*ca*sa*lambda2BL*xBL^2*H*H2]

LTot (:=)
LZpKin + LChi + LBLCurrent + LNuYuk + LScalarDiag

```

The last failure the loop worked on

This is the validation report handed to the repair agent at the start of `repair2/round3` — the problem it was asked to fix. The model **passed** after this round; the report below is the state that was repaired, not the final one.

Validation report — model.fr FAILED the tool chain

- FeynRules UFO compile: OK
- Hermiticity check: True
- Kinetic-terms check: True
- Mass-spectrum check: True
- MadGraph import: False
- Heuristic error tags: `mg5_import_fail`, `mg5_python_traceback`

FeynRules check hermiticity output

```

Checking for hermiticity by calculating the Feynman rules contained in L-HC[L].
If the lagrangian is hermitian, then the number of vertices should be zero.
Starting Feynman rule calculation.
Expanding the Lagrangian...
Expanding the indices over 8 cores
Collecting the different structures that enter the vertex.
No vertices found.
0 vertices obtained.
The lagrangian is hermitian.

```

FeynRules check kinetic_terms output

```

Neglecting all terms with more than 2 particles.
All kinetic terms are diagonal.

```

FeynRules check mass_spectrum output

```

Neglecting all terms with more than 2 particles.
All mass terms are diagonal.
Getting mass spectrum.
Checking for less then 0.1% agreement with model file values.

```

FeynRules / Wolfram Engine output (tail)

```

ctrum
[INFO] UFO output: /tmp/repair_bench2/B-L-SM/round2/UFO (AddDecays -> False)
--- Universal FeynRules Output (UFO) v 1.1 ---
Starting Feynman rule calculation.
Expanding the Lagrangian...
Expanding the indices over 8 cores

```

Collecting the different structures that enter the vertex.
 162 possible non-zero vertices have been found -> starting the computation: Dynamic[FR\$FeynmanRules] / 16
 157 vertices obtained.
 Flavor expansion of the vertices distributed over 8 cores: Dynamic[FR\$Count1] / 157
 - Saved vertices in InterfaceRun[1].

QN::NonConserv: Warning: non quantum number conserving vertex encountered!
 Quantum number LeptonNumber not conserved in vertex {GP, N1bar, e}.

QN::NonConserv: Warning: non quantum number conserving vertex encountered!
 Quantum number LeptonNumber not conserved in vertex {GP, N2bar, e}.

QN::NonConserv: Warning: non quantum number conserving vertex encountered!

General::stop: Further output of QN::NonConserv will be suppressed during this calculation.

Quantum number LeptonNumber not conserved in vertex {GP, N3bar, e}.
 Quantum number LeptonNumber not conserved in vertex {GP, N1bar, mu}.
 Quantum number LeptonNumber not conserved in vertex {GP, N2bar, mu}.
 Quantum number LeptonNumber not conserved in vertex {GP, N3bar, mu}.
 Quantum number LeptonNumber not conserved in vertex {GP, N1bar, ta}.
 Quantum number LeptonNumber not conserved in vertex {GP, N2bar, ta}.
 Quantum number LeptonNumber not conserved in vertex {GP, N3bar, ta}.
 Quantum number LeptonNumber not conserved in vertex {G0, N1bar, ve}.
 Quantum number LeptonNumber not conserved in vertex {H, N1bar, ve}.
 Quantum number LeptonNumber not conserved in vertex {G0, N2bar, ve}.
 Quantum number LeptonNumber not conserved in vertex {H, N2bar, ve}.
 Quantum number LeptonNumber not conserved in vertex {G0, N3bar, ve}.
 Quantum number LeptonNumber not conserved in vertex {H, N3bar, ve}.
 Quantum number LeptonNumber not conserved in vertex {G0, N1bar, vm}.
 Quantum number LeptonNumber not conserved in vertex {H, N1bar, vm}.
 Quantum number LeptonNumber not conserved in vertex {G0, N2bar, vm}.
 Quantum number LeptonNumber not conserved in vertex {H, N2bar, vm}.
 Quantum number LeptonNumber not conserved in vertex {G0, N3bar, vm}.
 Quantum number LeptonNumber not conserved in vertex {H, N3bar, vm}.
 Quantum number LeptonNumber not conserved in vertex {G0, N1bar, vt}.
 Quantum number LeptonNumber not conserved in vertex {H, N1bar, vt}.
 Quantum number LeptonNumber not conserved in vertex {G0, N2bar, vt}.
 Quantum number LeptonNumber not conserved in vertex {H, N2bar, vt}.
 Quantum number LeptonNumber not conserved in vertex {G0, N3bar, vt}.
 Quantum number LeptonNumber not conserved in vertex {H, N3bar, vt}.
 Quantum number LeptonNumber not conserved in vertex {GPbar, ebar, N1}.
 Quantum number LeptonNumber not conserved in vertex {GPbar, ebar, N2}.
 Quantum number LeptonNumber not conserved in vertex {GPbar, ebar, N3}.
 Quantum number LeptonNumber not conserved in vertex {GPbar, mubar, N1}.
 Quantum number LeptonNumber not conserved in vertex {GPbar, mubar, N2}.
 Quantum number LeptonNumber not conserved in vertex {GPbar, mubar, N3}.
 Quantum number LeptonNumber not conserved in vertex {GPbar, tabar, N1}.
 Quantum number LeptonNumber not conserved in vertex {GPbar, tabar, N2}.
 Quantum number LeptonNumber not conserved in vertex {GPbar, tabar, N3}.
 Quantum number LeptonNumber not conserved in vertex {G0, vebar, N1}.
 Quantum number LeptonNumber not conserved in vertex {H, vebar, N1}.
 Quantum number LeptonNumber not conserved in vertex {G0, vebar, N2}.
 Quantum number LeptonNumber not conserved in vertex {H, vebar, N2}.
 Quantum number LeptonNumber not conserved in vertex {G0, vebar, N3}.
 Quantum number LeptonNumber not conserved in vertex {H, vebar, N3}.

Quantum number LeptonNumber not conserved in vertex {G0, vmbar, N1}.

Quantum number LeptonNumber not conserved in vertex {H, vmbar, N1}.

Quantum number LeptonNumber not conserved in vertex {G0, vmbar, N2}.

Quantum number LeptonNumber not conserved in vertex {H, vmbar, N2}.

Quantum number LeptonNumber not conserved in vertex {G0, vmbar, N3}.

Quantum number LeptonNumber not conserved in vertex {H, vmbar, N3}.

Quantum number LeptonNumber not conserved in vertex {G0, vtbar, N1}.

Quantum number LeptonNumber not conserved in vertex {H, vtbar, N1}.

Quantum number LeptonNumber not conserved in vertex {G0, vtbar, N2}.

Quantum number LeptonNumber not conserved in vertex {H, vtbar, N2}.

Quantum number LeptonNumber not conserved in vertex {G0, vtbar, N3}.

Quantum number LeptonNumber not conserved in vertex {H, vtbar, N3}.

Preparing Python output.

- Splitting vertices into building blocks.

Splitting of vertices distributed over 8 kernels.

- Optimizing: Dynamic[PRIVATE`PY\$SplitVertexCounter]/201 .
- Writing files.

Warning: Non positive interaction order QED.

This might reduce the efficiency of certain matrix element generators.
See logfile for more details.

Done!

[INFO] Done.

MadGraph import output (tail)

```
aMC/models/import_ufo.py", line 251, in import_model
    model = import_full_model(model_path, decay, prefix)
File "$MG5_PATH/models/import_ufo.py", line 413, in import_full_model
    ufo_model = ufomodels.load_model(model_path, decay)
File "$MG5_PATH/models/__init__.py", line 101, in load_model
    raise UFOError(str(error))
models.UFOError: invalid syntax (object_library.py, line 268)
Command "import /private/tmp/repair_bench2/B-L-SM/round2/mg5run/mg5_import.txt" interrupted in sub-command
"import model /tmp/repair_bench2/B-L-SM/round2/UFO" with error:
UFOError : invalid syntax (object_library.py, line 268)
Please report this bug on https://bugs.launchpad.net/mg5amcnlo
More information is found in 'MG5_debug'.
Please attach this file to your report.
```

MG5_debug (the real MadGraph error)

```
5_aMC/models/import_ufo.py", line 611, in load_model
    self.add_interaction(interaction_info, color_info)
~~~~~
File "$MG5_PATH/models/import_ufo.py", line 1840, in add_interaction
    raise InvalidModel('Some couplings have \'1\' order.
    This is not allowed in MG.
    Please defines an additional coupling to your model')
models.import_ufo.InvalidModel: Some couplings have '1' order.
    This is not allowed in MG.
    Please defines an additional coupling to your model
```

During handling of the above exception, another exception occurred:

Traceback (most recent call last):

```

File "$MG5_PATH/madgraph/interface/extended_cmd.py", line 1570, in onecmd
    return self.onecmd_orig(line, **opt)
~~~~~
File "$MG5_PATH/madgraph/interface/extended_cmd.py", line 1519, in onecmd_orig
    return func(arg, **opt)
File "$MG5_PATH/madgraph/interface/master_interface.py", line 281, in do_import
    self.cmd.do_import(self, *args, **opts)
~~~~~
File "$MG5_PATH/madgraph/interface/madgraph_interface.py", line 5893, in do_import
    self.import_command_file(args[1])
~~~~~
File "$MG5_PATH/madgraph/interface/extended_cmd.py", line 1718, in import_command_file
    self.exec_cmd(line, precmd=True)
~~~~~
File "$MG5_PATH/madgraph/interface/extended_cmd.py", line 1599, in exec_cmd
    stop = Cmd.onecmd_orig(current_interface, line, **opt)
File "$MG5_PATH/madgraph/interface/extended_cmd.py", line 1519, in onecmd_orig
    return func(arg, **opt)
File "$MG5_PATH/madgraph/interface/master_interface.py", line 281, in do_import
    self.cmd.do_import(self, *args, **opts)
~~~~~
File "$MG5_PATH/madgraph/interface/madgraph_interface.py", line 5835, in do_import
    raise err
File "$MG5_PATH/madgraph/interface/madgraph_interface.py", line 5819, in do_import
    self._curr_model = import_ufo.import_model(args[1], prefix=prefix,
~~~~~
                        complex_mass_scheme=self.options['complex_mass_scheme'],
                        ~~~~~
                        options=options)
                        ~~~~~
File "$MG5_PATH/models/import_ufo.py", line 251, in import_model
    model = import_full_model(model_path, decay, prefix)
File "$MG5_PATH/models/import_ufo.py", line 413, in import_full_model
    ufo_model = ufomodels.load_model(model_path, decay)
File "$MG5_PATH/models/__init__.py", line 101, in load_model
    raise UFOError(str(error))
models.UFOError: invalid syntax (object_library.py, line 268)
Fail to write options with error No model currently active, please import a model!

```

Repair history

What the loop tried, and what each attempt measured.

Repair history — previous attempts on this model

Attempt 1

What was changed (the agent's own summary): Changed `model.fr` in place by renaming the internal Yukawa tensors `yM` and `ynu` to `YMR` and `YNU` throughout their parameter declarations and neutrino Yukawa terms, which should prevent FeynRules from exporting malformed UFO parameter names like `yM1x2 /. yM`. I also added `LScalarDiag` just before the final `LTot` assignment and included it in `LTot` to cancel the reported residual H/H_2 kinetic and mass mixing terms, keeping the declared scalar mass basis diagonal. I could not run the full validation locally because `wolframscript` cannot find a Wolfram kernel here and `MadGraph` is not on `PATH`.

Validation outcome AFTER that change: `status=compiled, hermiticity=True, kinetic=True, mass=True, madgraph_import=False, tags=mg5_import_fail, ufo_python_syntax_error, mg5_python_traceback,`

mg5__invalid__cmd, compile_seconds=76.2

Attempt 2

What was changed (the agent's own summary): Changed `model.fr` to replace the problematic internal indexed Yukawa parameters `YMR` and `YNU` with explicit scalar internal component parameters (`YMR1x1 ... YMR3x3`, `YNU1x1 ... YNU3x3`) carrying the same formulas and zero entries, plus explicit NP interaction order. I then rewired `LNuYukNonHC` through local replacement rules so the operator structure is unchanged but `FeynRules` exports valid UFO identifiers instead of malformed Mathematica `ReplaceAll` names like `YMR1x2 /. YMR`. `LTot` remains the final top-level assignment. I did not run the full Wolfram/MadGraph validation because that would require tools outside the provided directory.

Validation outcome AFTER that change: `status=compiled`, `hermiticity=True`, `kinetic=True`, `mass=True`, `madgraph_import=False`, `tags=mg5__import__fail`, `mg5__python__traceback`, `compile_seconds=76.0`

For reference — the ORIGINAL model's outcome before any repair:

`status=compiled`, `hermiticity=True`, `kinetic=True`, `mass=True`, `madgraph_import=False`, `tags=mg5__import__fail`, `ufo_python_syntax_error`, `mg5__python__traceback`, `mg5__invalid__cmd`, `compile_seconds=67.4`

Physicist sign-off

- Reviewed by: _____ Date: _____
- Verdict: ☐ approve ☐ approve with corrections ☐ reject
- Notes: